Trustworthy Collective Reasoning and Evaluation for AI Agents

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1 The problem

AI agents built on large language models (LLMs) increasingly evaluate one another and decide together: they judge one another's answers, peer-review papers, supply the rewards that train new models, and vote or discuss (Zheng et al., 2023; Luo et al., 2025; Lee et al., 2024). Yet a single LLM judge has known biases, such as favoring longer answers or answers like its own (Zheng et al., 2023; Panickssery et al., 2024). Collective reasoning can do better than a single judge (Verga et al., 2024; Du et al., 2024), but averaging or voting stays biased when the judges share mistakes. **The central problem of this project is how a group of AI agents of unknown and correlated reliability, some possibly dishonest, can reach trustworthy decisions and evaluations.**

For example, five AI agents read shipping reports and are asked whether a cargo vessel has left port. Three say yes and two say no, but the vessel is still in port. Two of the yes answers rest on the same outdated schedule, and the third comes from a compromised agent that argues confidently for yes. A majority vote picks the wrong answer, and open discussion may do no better, because the compromised agent is persuasive and the two agents with the same source back each other up.

Verifying each claim directly, by an analyst or a satellite image, would avoid such errors but is slow and costly, and AI agents produce claims faster than people can verify them. We propose debate as a foundation for trustworthy evaluation by AI: two agents argue for opposite answers, exposing each other's flaws, until the dispute narrows to one step a judge can verify (Irving et al., 2018). To work among many AI agents and let them learn from it, debate must overcome the example's three failures: agents that share a mistake seem to confirm each other, a dishonest agent can argue, rate others, or judge strategically, and a judge that sees only words may favor the more persuasive side.

2 Why it matters

All three failures have been measured. Models' errors are correlated, more so as they improve (Kim et al., 2025; Goel et al., 2025), so agreement is weaker evidence than it looks: in one study, nine AI judges were worth about two independent ones (Kohli, 2026). Agents collude: pairs verifying each other's work colluded in 94% of runs when cheating paid more (Shi et al., 2026). Persuasion misleads: one persuasive agent can sharply cut a group's accuracy (Amayuelas et al., 2024).

Navy decisions face all of these problems at once: AI tools differ in reliability that is rarely known in advance, the true answer often arrives late, an adversary may compromise an agent or forge reports, and verification is costly or impossible when connectivity is limited. A commander must still decide whether to act on what the agents agree on.

3 How it is solved today, and the limitations

Statistical methods estimate each rater's accuracy without an answer key (Dawid and Skene, 1979; Whitehill et al., 2009) and detect sources that copy one another (Dong et al., 2009), but they assume that raters err independently or in simple patterns (Jaffe et al., 2016) and never change their minds.

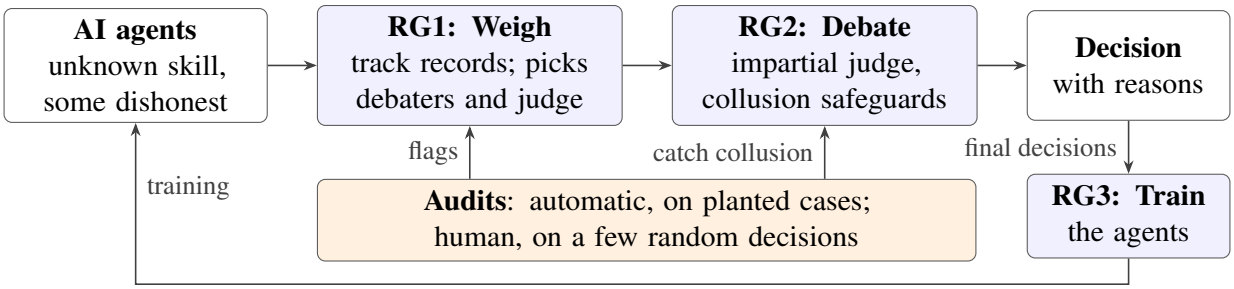


Figure 1: RG1 weighs answers and picks debaters and a judge. RG2 replaces open discussion with debate before an impartial judge, and audits catch collusion. RG3 trains the agents on the group's decisions.

Computational social choice designs impartial peer review, in which a participant's ratings cannot change its own chance of selection (Kurokawa et al., 2015; Alon et al., 2011), and fair panel selection (Flanigan et al., 2021). Lev et al. (2023) added reliability weights to impartial selection, but these reward agreeing with the majority, which fails once accurate reviewers are a minority. These mechanisms take opinions as given, whereas in our setting some claims can be verified.

Prior work on LLM discussion (Du et al., 2024; Li et al., 2024a) is limited: majority voting explains most of its gain (Choi et al., 2025), and it assumes honest agents that exchange only text. Standard debate helps weaker judges (Irving et al., 2018; Khan et al., 2024), but trusts the judge, assumes the sides truly compete, and grounds claims at most in quoted text. No existing debate protocol holds up when agents share mistakes, lie, or collude. Nor can agents yet keep improving at scale without an answer key: rewarding majority agreement collapsed accuracy (Shafayat et al., 2025), rewarding debate wins taught agents to invent quotes (Kenton et al., 2026), and rewarding approval taught them to mislead people (Wen et al., 2025).

4 Our approach: what is new, and why it will succeed

We propose a framework in which groups of AI agents reach trustworthy decisions and evaluations through debate (Figure 1). Each agent first answers alone, with its evidence. The project has three interconnected research goals (RGs). RG1 weighs each answer by its agent's track record, counting shared mistakes once, and picks debaters and judges so that no agent can favor itself. RG2 settles disputes by agentic debate instead of open discussion: if no answer clearly leads, two representatives debate before an impartial judge, and the judge's choice is the group's decision, so the agents need not agree. Any tool evidence must be re-runnable, and automatic and human audits catch debaters or judges who collude. RG3 lets the agents improve themselves: reinforcement learning on the group's decisions credits each message for helping reach the decision rather than for winning, needs no human labels, and keeps working after the agents outgrow human judges.

RG1. Decide what to debate, and pick debaters and judges impartially

A plain vote lets shared mistakes pile up, and agents choosing debaters can favor themselves.

Task 1.1: Deciding what to debate, counting shared mistakes once. Each answer counts according to its agent's track record, estimated from who agrees with whom across real questions, as one can grade an exam without the answer key; agents flagged by audits (Task 2.2) count less. The core method extends the rater model of Dawid and Skene (1979) with shared error factors: agents that keep giving the same unusual answers share a factor, and their agreement adds little

evidence. In the vessel example, the two agents sharing an outdated schedule count about once, so yes no longer leads; with no clear leader, the question goes to debate.

Task 1.2: Impartial choice of debaters and judges. A debate needs a judge and one representative per answer. Representatives are chosen by peer ratings of their arguments, with raters weighted by track record, as in our PeerEval leaderboard and PRD method (Li et al., 2024a). So that no agent can rate its way into a role, we adapt the partition idea of impartial peer review (Alon et al., 2011; Kurokawa et al., 2015): agents sharing an underlying model form a group, rated only by outsiders. Each answer's representative is the top-rated member of a randomly drawn group that gave that answer; the judge is a random agent that did not answer and shares no model with either side. Ratings pick only who argues; the judge picks the answer.

RG2. Settle disputes by agentic debate, and catch collusion

Open discussion erases disagreement and rewards persuasion, and debaters can collude.

Task 2.1: Debate before an impartial judge. Today's multi-agent systems discuss until they agree, which erases the disagreement that exposes errors (Shu, 2026). We instead run debate in a harness, the software around models that turns them into agents: a sandbox with tools, a control loop that decides what happens next, memory, and verification of results (Ning et al., 2026). Our control loop sets who speaks, who judges, and when to stop: Task 1.2 picks one representative per answer and a judge with no stake in either, the two walk through their arguments to the first step on which they differ (Irving et al., 2018), and the loop stops there for the judge. The judge's choice is the group's decision; most LLM discussions run fixed rounds of text between given models. Tools are optional, but tool evidence must carry its exact call, which the judge, itself an agent (Zhuge et al., 2025), re-runs in the sandbox. This builds on PRD's peer discussion and HiPRAG's search agents, and the harness logs each debate for audits and training. In the vessel example, the no side's query returns a newer port report listing the vessel at its berth; the judge re-runs it and picks no.

Task 2.2: Automatic and human audits against collusion. Collusion here means debaters, or the judge, steering a debate to the wrong side. A debater can argue its side badly on purpose, or colluders can mislead the judge with true but selectively chosen facts (Hu et al., 2026), which re-running evidence does not catch. Automatic audits mix in planted debates on planted cases, kept out of training, such as a paper with a deliberately inserted error, as in studies of human peer review (Schroter et al., 2008). Debaters and judges whose planted debates keep ending on the wrong side are flagged. Because capable models can often tell evaluations from real use (Needham et al., 2025), a colluder could behave well only in planted debates. Human audits therefore cover a few real decisions chosen at random: a person (such as an analyst) or a direct measurement (such as a satellite image) finds the right answer. As with random tax audits, no agent knows which decisions will be audited, and agents that do worse on audited decisions than on planted ones are flagged.

RG3. Let agents self-improve with reinforcement learning on the group's decisions

Self-improvement needs rewards that track correct answers, not votes or debate wins.

Task 3: Rewards for contributions to trustworthy decisions. We will train the agents with reinforcement learning on the group's final decisions, which need no human labels and are harder to game than a majority vote or a debate win, because shared mistakes count once and the judge re-runs tool evidence. Each message is rewarded for moving the group toward the decision; tool-backed

decisions weigh more, and decisions involving flagged agents are dropped. This extends our process reward models, FG-PRM and HiPRAG (Li et al., 2025; Wu et al., 2026), from one agent's steps to many agents' messages. The main experiment compares this reward with rewards for majority agreement and for debate wins (Arnesen et al., 2024).

Evaluation: testbeds and a red team

All data are synthetic or public. We will use (A) math and code problems solved by agents of known skill, some sharing a flawed hint; (B) multi-document questions from our MEQA pipeline (Li et al., 2024b), including synthetic maritime scenarios; and (C) scientific claims and peer review, with reproducibility studies as audits. Following AI control (Greenblatt et al., 2024), a red team will train debaters and judges to collude, to spot planted debates, and to cheat only when they expect no audit, and we will measure how many wrong decisions get through at each audit budget. Baselines are a single AI judge, majority vote, and averaging with equal weights.

What is new. (i) Weights that count shared mistakes about once, and impartial choice of debaters and judges. (ii) In place of open discussion, a debate harness that stops at the disputed step for an impartial judge, who can re-run tool evidence. (iii) Automatic audits on planted debates and a few human audits that expose colluding debaters or judges, even those behaving well only in planted debates. (iv) Self-improvement by reinforcement learning on the group's decisions, with each message credited for helping reach them.

Why it will succeed. Theory supports the design: in debate, a judge who can verify one disputed step can settle arguments far beyond its own ability (Irving et al., 2018), and our impartial judge rules on that step. Without trusted audits, dishonest agents that imitate honest ones cannot be told apart from them, but a number of audits that does not grow with the number of items suffices, even when most agents are dishonest (Steinhardt et al., 2016). Most tasks build on the PI's own previous work. For example, PRD agrees better with human judgments than the strongest single AI judge (Li et al., 2024a), and HiPRAG, our method for agentic search that grounds answers in retrieved evidence, cuts unnecessary searches to 2.3% while missing fewer needed ones (Wu et al., 2026). The PI built PeerEval, the first leaderboard on which AI models rank one another, co-organized the AI4Research workshop at AAAI 2025, and co-wrote a survey of LLMs for scientific research, including peer review (Luo et al., 2025). The PI also co-led one of five red teams in the Amazon Nova AI Challenge, holds a 2024 NSF CAREER award and a 2023 Amazon Research Award, and as a postdoctoral researcher led system integration for RESIN-11 in DARPA KAIROS.

Limitations. Where no tool applies, the judge rules on argument alone, as in standard debate, and Navy use would need Navy data and expert validation.

5 Impact if successful

The project will show when a debate among AI agents can be trusted and how few audits it takes to keep the agents honest. Because people audit only a few decisions and training needs no human labels, the methods are built to scale to superhuman AI, whose answers and debates people cannot easily evaluate, helping keep it honest and safe. A Navy analyst using several AI tools would get a decision with its reasons: whose judgments carried weight and why, which step decided the debate, and which evidence the judge reproduced. These methods would let conferences and funders try impartial, audited peer review by AI. We will release code, testbeds, and protocols.

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